Chapter 11, Problem 17PP

Problem

The monthly reading of a paper mill's meter is as follows:

Maximum demand: 48,000 kW

Energy consumed: 750 MWh

Using the two-part rate in Example 11.17, calculate the monthly bill for the paper mill.

Example 11.17:

A manufacturing industry consumes 200 MWh in one month. If the maximum demand is 1,600 kW, calculate the electricity bill based on the following two-part rate:

Demand charge: $5.00 per month per kW of billing demand.

Energy charge: 8 cents per kWh for the first 50,000 kWh, 5 cents per kWh for the remaining energy.

Solution:

The demand charge is

$$\$5.00 \times 1{,}600 = \$8{,}000 \quad \textbf{(11.17.1)}$$

The energy charge for the first 50,000 kWh is

$$\$0.08 \times 50{,}000 = \$4{,}000 \quad \textbf{(11.17.2)}$$

The remaining energy is 200,000 kWh − 50,000 kWh = 150,000 kWh, and the corresponding energy charge is

$$\$0.05 \times 150{,}000 = \$7{,}500 \quad \textbf{(11.17.3)}$$

Adding the results of Eqs. (11.17.1) to (11.17.3) gives

Total bill for the month = $8,000 + $4,000 + $7,500 = $19,500

It may appear that the cost of electricity is too high. But this is often a small fraction of the overall cost of production of the goods manufactured or the selling price of the finished product.

Step-by-step solution

Step 1 of 4

Calculate the value of the demand charge.

$$\begin{aligned} C_1 &= 5 \times 48{,}000 \\ &= \$\ 240{,}000 \end{aligned}$$

Therefore, the value of the demand charge C_1 is $\$\ 240{,}000$.

Step 2 of 4

Calculate the energy charge for the first $50{,}000\ \text{kWh}$ units.

$$\begin{aligned} C_2 &= 0.08 \times 50{,}000 \\ &= \$4{,}000 \end{aligned}$$

Therefore, the value of the energy charge C_2 for the first $50000\ \text{kWh}$ units is $\$4{,}000$.

Step 3 of 4

Calculate the energy charge for the remaining units.

$$\begin{aligned} C_3 &= 0.05 \times 700{,}000 \\ &= \$35{,}000 \end{aligned}$$

Therefore, the value of the energy charge C_3 for remaining units is $\$35{,}000$.

Step 4 of 4

Calculate the value of the total cost.

$$C = C_1 + C_2 + C_3$$

Substitute $\$\ 240{,}000$ for C_1, $\$4{,}000$ for C_2 and $\$35{,}000$ for C_3.

$$\begin{aligned} C &= 240{,}000 + 4{,}000 + 35{,}000 \\ &= \$\ 279{,}000 \end{aligned}$$

Therefore, the value of the total cost C is $\boxed{\$\ 279{,}000}$.